

ASTR 210 Fall 2025 — Homework set 13
50 points and optional extra credit
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Solutions need to show all important intermediate calculations, diagrams, and explanations in addition to the final answer to receive full credit. Please check the legibility of homework solutions uploaded to the class website as we cannot grade or give credit for illegible or unreadable work. Units do not need to be specified if an expression or equation is requested; for numerical calculations please use the specified units for the answer, or if not specified, SI or cgs units. Please use Table A.1 for the value of physical constants and Table A.2 for the value of astronomical constants, unless otherwise specified in the problem. Please use the correct number of significant figures.

13.1 Cosmology missing steps

A lot of cosmology is basically math, so we'll do a bunch of math. Write out the missing steps that connect the following equations in your text.

- (a) Begin with equation 23.22. Make the substitutions as noted and do the manipulations to arrive at equation 23.23. [3 pts]
- (b) Similarly, begin with 23.23 and show the missing steps to arrive at 23.24. [3 pts]
- (c) ... and 23.25 [3 pts]
- (d) ... and 23.26 [2 pts]
- (e) From 23.26, show that CMB photons redshift according to $\lambda(t) \propto a(t)$ as described in the paragraph after equation 23.26. You may assume that the CMB spectrum retains a blackbody shape throughout the expansion of the universe, and you are interested in a "typical" photon at the peak of the blackbody spectrum. [4 pts]
- (f) Begin with equation 23.28 and show the missing work to arrive at equation 23.32. [4 pts]
- (g) ... and 23.71 to 23.73. [2 pts]
- (h) Show the missing work to get from equation 23.71 (a.k.a. 24.11) to 24.14 (a.k.a. 24.35). [3 pts]
- (i) ... and 24.35 to 24.36. [4 pts]

13.2 Precision cosmological measurements

The most distant galaxy with a distance estimate based on pure geometric methods (rather than a ladder-based estimate like variable stars or supernovae) is NGC 5765b, at 112 Mpc.¹ Its distance and its recession velocity can be used to calculate the Hubble constant, of course. Unfortunately for us, different methods for calculating H_0 are now starting to give slightly different results. We'll try some calculations to see whether those results are significantly different.

- (a) Two recent estimates of H_0 are $67 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $74 \text{ km s}^{-1} \text{ Mpc}^{-1}$. For an object at a distance of 112 Mpc, what is the redshifted Doppler velocity that would be expected for those two values of H_0 ? [4 pts]
- (b) This geometric method uses observations of a specific spectral line from water molecules. The rest frequency of the line in question is $\nu = 22.23508 \text{ GHz}$. Calculate the wavelength of the spectral line. [2 pts]

¹<https://iopscience.iop.org/article/10.3847/2041-8213/ab75f0>, if you are really interested.

- (c) If you want to be able to distinguish between those two values of the Hubble constant mentioned above, what is the precision you need on your wavelength observations? In other words, calculate the redshifted wavelength you would expect to observe in the two cases and note their difference. [4 pts]

13.3 Matter–dark energy equality

Your text (e.g. Table 24.1) adopts a consensus cosmological model which is flat, with $\Omega_m = 0.30$ and $\Omega_\Lambda = 0.70$. Cosmologists are continually tweaking these estimates upon receipt of new data, of course. For the purposes of this problem, let's explore a flat cosmological model with $\Omega_m = 0.27$ and $\Omega_\Lambda = 0.73$. Ignore the contribution of radiation to the energy density of the universe.

- (a) For these parameters, calculate the redshift of equality (z_{eq}) between matter (m) and dark energy (Λ). [4 pts]
- (b) Calculate the acceleration $\ddot{a}$ at redshift z_{eq} , in terms of the Hubble constant H_0 . Was the universe's expansion accelerating or decelerating at that time? [4 pts]
- (c) Calculate the value of the Hubble parameter $H(t)$ at the redshift z_{eq} , in terms of its current value H_0 . [4 pts]

EXTRA CREDIT

13.4 Curvature

Suppose that we flatten out all the bumps on the surface of Earth, including the equatorial bulge, until it is a perfect sphere of radius 6371 km. If we mark out an equilateral triangle with sides of length $L = 1$ km, what will be the sum of the interior angles in our triangle? [5 pts]

13.5 Proper distances

Continued from problem 13.3 above. Estimate the current proper distance to an object at redshift z_{eq} . For this calculation, assume that the scale factor evolves linearly with time between z_{eq} and today, so that $a \propto t^1$, and assume that $t_0 = (H_0)^{-1}$. Compare that current proper distance to the value you would get from a simple application of the Hubble law. Assume the current value is $H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$. *Note: this last part is in the material that has been excluded from the syllabus, which is why it's part of an extra credit problem rather than a regular homework problem. It's still excluded from the final exam. But it's fun to see how these calculations are done.* [5 pts]